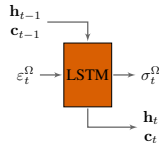
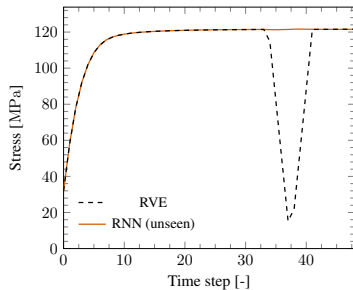


# Machine learning to the rescue?

Popular solutions (e.g. RNNs) can be tricky to train:

- Extensive training set needed to guarantee generalization
- Sensitivity to time step size, stopping, etc



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Relearning basic physics has a number of drawbacks:

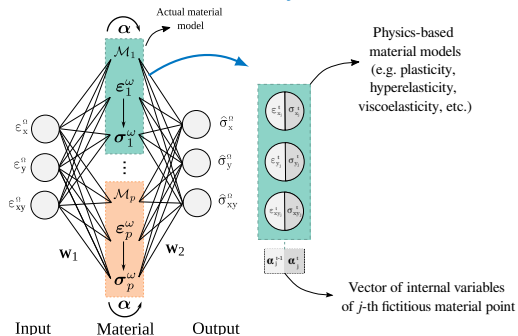
- Ensuring solution robustness is challenging (negative tangent, spurious localization, etc)
- Non-monotonic strain paths cannot be treated trivially

# Hybrid models

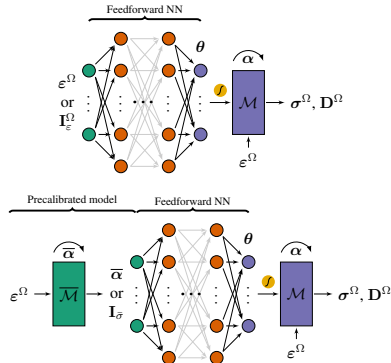
Embedding **intact** material models in ML architectures:

- Natural choice of models to embed: **same as in microscale!**
- Physically recurrent, ML does not touch path dependency
- **This presentation:** single material model, evolving material properties

Marina Maia, Thursday 10:00, Room B!



[Maia et al. (2023), *CMAME*, **407**:115934]



[Rocha et al. (2023), *Mech Mater*, **184**:104707]

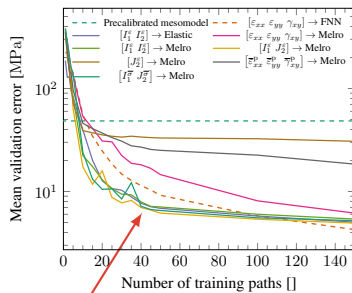
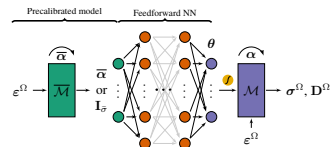
# Plasticity decoder

[Melro et al. (2013), *Int J Sol Struc*, **50**:1897-1905]

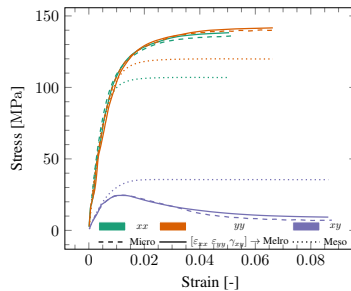
[Vogler et al. (2013), *Mech Mater*, **59**:50-64]

Elastoplastic decoder by Melro et al, same as for microscale resin:

- Yield stresses and plastic Poisson ratio controlled by the network
- Trying out different combinations of input features
- Comparing with pre-calibrated mesomodel by Vogler et al



6800 trained nets,  $\approx 1$  million test models



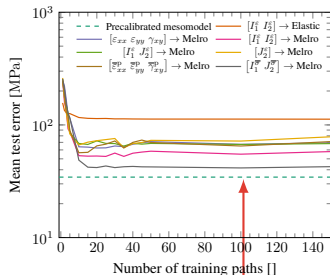
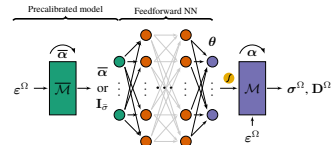
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